

9	D	$F = ma = m \left(\frac{v - u}{t} \right) = \frac{mv}{t} \text{ since } u = 0$ $\text{K.E.} = \frac{1}{2}mv^2 = \frac{(mv)^2}{2m} = \frac{F^2 t^2}{2m} \Rightarrow \text{K.E.} \propto \frac{t^2}{m}$ $\frac{\text{gain in kinetic energy of body A}}{\text{gain in kinetic energy of body B}} = \frac{\text{K.E.}_A - 0}{\text{K.E.}_B - 0}$ $= \frac{t_A^2}{m_A} \div \frac{t_B^2}{m_B}$ $= \frac{t_A^2 m_B}{t_B^2 m_A}$ $= \left(\frac{2t}{t} \right)^2 \left(\frac{2m}{m} \right)$ $= 8$

Raffles Institution
Year 5-6 Physics Department

10	D	As the sphere moves from highest to lowest point
----	---	--